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REQUIREMENT SPECIFICATION & DESIGN DOCUMENT

STUDENT MANAGEMENT SYSTEM

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CHAPTER 1: FUNCTIONAL REQUIREMENTS

The Student Management System shall support academic and administrative operations for three user roles: Administrator, Student, and Lecturer. The system shall restrict access to features and data according to the role of each user

1. Authentication and Authorization

Registration: The system shall allow only the Administrator to create student and lecturer accounts based on official admission records or personnel information. The system shall require the following fields: automatically generated Student ID or Lecturer ID, Citizen ID Number, Full Name, Date of Birth, Email, Gender, Phone Number, and Faculty/Major. The system shall validate all inputs upon submission. The system shall reject the creation request and display an error message identifying the invalid field if any mandatory field is missing or any unique field already exists. Upon successful registration, the system shall automatically generate an account with default credentials and create the corresponding student record or lecturer profile and notify the user

Log in: The system shall allow registered users to select a login role (Administrator, Student or Lecturer) and log in using a username and password. After successful authentication, the system shall identify the user's role and display the interface and functions permitted for that role. If the credentials are invalid or the account is locked, the system shall reject the request and display an error message stating the reason

Log out: The system shall allow users to log out at any time. Upon logout, the system shall terminate the active session and clear all temporary session data

Password Change: The system shall allow students and lecturers to change their passwords by entering the current password and a new password. The system shall require the new password to meet the password complexity rules and to be confirmed by the user before saving the change

Password Recovery: The system shall allow users to recover their accounts by providing a registered email address or phone number. After successfully verifying account ownership, the system shall allow the user to set a new password

2. User Management

Student Management: The system shall allow the Administrator to view, update, and delete student records. The system shall require each student record to have an academic or administrative status. The system shall allow students to view and update their own contact information. The system shall not allow students to modify their identification data or affiliated faculty

Lecturer Management: The system shall allow the Administrator to view, update, and delete lecturer profiles and to assign lecturers to class sections. The system shall allow lecturers to view and update their own contact information. The system shall not allow lecturers to modify their identification data or affiliated faculty

3. Academic Management

Curriculum Management: The system shall allow students to view the standard curriculum of their major, including compulsory courses, elective courses, credit values, and prerequisites. The system shall automatically mark the completion status of each course based on the student's academic transcript

Semester Management: The system shall allow the Administrator to create, view, configure and delete academic semesters, including Semester Name, Academic Year, Start Dates and End Dates. The system shall link all academic activities and class section offerings to a specific semester. The system shall reject the deletion of a semester that is linked to class sections or historical academic records

Faculty Management: The system shall allow the Administrator to create, view, update, and delete faculty records, each containing Faculty Code, Faculty Name, Email, and Phone Number. The system shall reject the deletion of any faculty that currently manages active courses or has enrolled students

Course Management: The system shall allow the Administrator to create, view, update, and delete courses. The system shall require each course record to include Course Code, Course Name, Credits, Managing Faculty, and Prerequisite Constraints.

The system shall reject the deletion of a course that is linked to ongoing class sections or historical academic records

Class Section Management: The system shall allow the Administrator to manage multiple class sections of a course within a specific semester. The system shall require each class section record to include Class Code, Course Code, Semester, Assigned Lecturer, Maximum Capacity, Current Enrollment, Room, and Class Schedule

4. Grade Management

Gradebook Management: The system shall allow lecturers to record student grades for midterms, assignments, attendance, and final examinations. The system shall reject any numerical mark outside the range of 0.0 to 10.0. The system shall automatically calculate the final course grade using the weighted grading formula defined for each course and assign the corresponding letter grade

GPA Calculation: The system shall automatically calculate each student's Semester GPA and Cumulative GPA based on finalized academic results and credit-weighted grading rules

Academic Transcript Viewing: The system shall allow students to view their academic history, including component marks, final exam results, and course letter grades. The system shall automatically update the Semester GPA and Cumulative GPA. The system shall display only finalized grades to students

CHAPTER 2: NON-FUNCTIONAL REQUIREMENTS

Security: The system shall protect user data from unauthorized access. The system shall automatically terminate an active session after 15 minutes of inactivity, clear all session data, and redirect the user to the login page

Usability: The system shall provide a consistent and responsive web interface across all functions. The system shall display error messages that identify the invalid field or the schedule conflict that caused the error. The system shall display a confirmation or error message for every user action, including successful updates, failed operations, and invalid submissions

Reliability: The system shall maintain data integrity and consistency during database operations. The system shall confirm completion of an operation only after all related record changes have been committed successfully. If a database operation fails, the system shall roll back all related changes, display an error message, and continue running without crashing

Performance: The system shall be designed for small-scale institutional deployment. Under normal operating conditions, the system shall respond to user actions within 15 seconds

Maintainability: The system shall follow a modular architecture to support testing, debugging, and extension. The source code shall include comments, follow consistent naming conventions, and comply with the PEP 8 coding standard. The system structure shall allow business rules to be modified and the data storage mechanism to be replaced without redesigning the entire system

Design Constraints: The system shall be a web-based application. The user interface shall be developed using HTML, CSS, and JavaScript. The core business logic shall be implemented in Python. The system shall use a Microsoft SQL Server Relational Database Management System (RDBMS) for data persistence. The system shall be deployed and hosted on a Windows 11 server environment

CHAPTER 3: DATA FLOW DIAGRAM

1. Overview of DFD

This document describes two levels of the Data Flow Diagram (DFD) of the Student Management System (SMS). A DFD is a modeling tool used to represent external entities, processing functions and the data flows exchanged between them

The two DFD levels included in this document are:

- **DFD Level 0 (Context Diagram):** A high-level overview that treats the entire system as a single process and shows its interactions with three external entities: Administrator, Student and Lecturer
- **DFD Level 1:** Decomposes the main process into four sub-processes, showing the detailed data flows between these sub-processes, the external entities and the four data stores through which the sub-processes read, write and share data

2. DFD Diagram Level 0 - Context Diagram

2.1. Overview

DFD Level 0, also known as the Context Diagram, represents the entire Student Management System as a single process labeled “0.0 - Student Management System”. The diagram defines the system boundary and shows the data flows exchanged between this process and three external entities: Administrator, Student and Lecturer. No internal processing is shown at this level

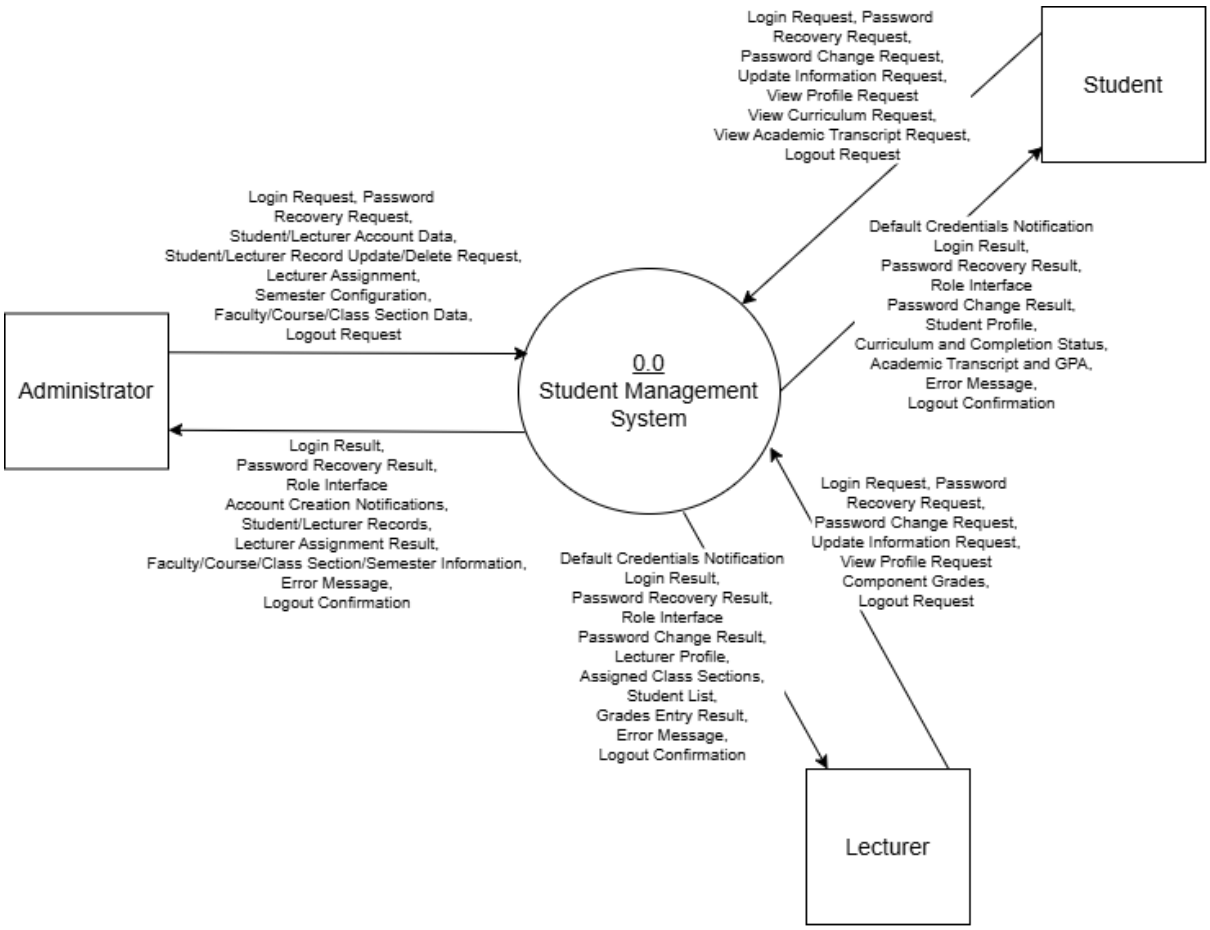


Image 1: DFD Diagram Level 0

2.2. External Entities

Entity	Role	Description
Administrator	System Administrator	Manages the entire system with the highest level of access, responsible for user management, system configuration and academic data management.
Student	End User	Uses the system to view academic transcripts and GPA, view the curriculum, check completion status, view and update personal information.

Lecturer	End User	Uses the system to enter component grades, view assigned class sections, student lists, view and update personal information.

3. DFD Diagram Level 1

3.1. Overview

DFD Level 1 decomposes the central process “0.0 - Student Management System” into four sub-processes, showing the detailed data flows between these sub-processes and the three external entities (Administrator, Student and Lecturer), as well as the four data stores through which the sub-processes read, write and share data

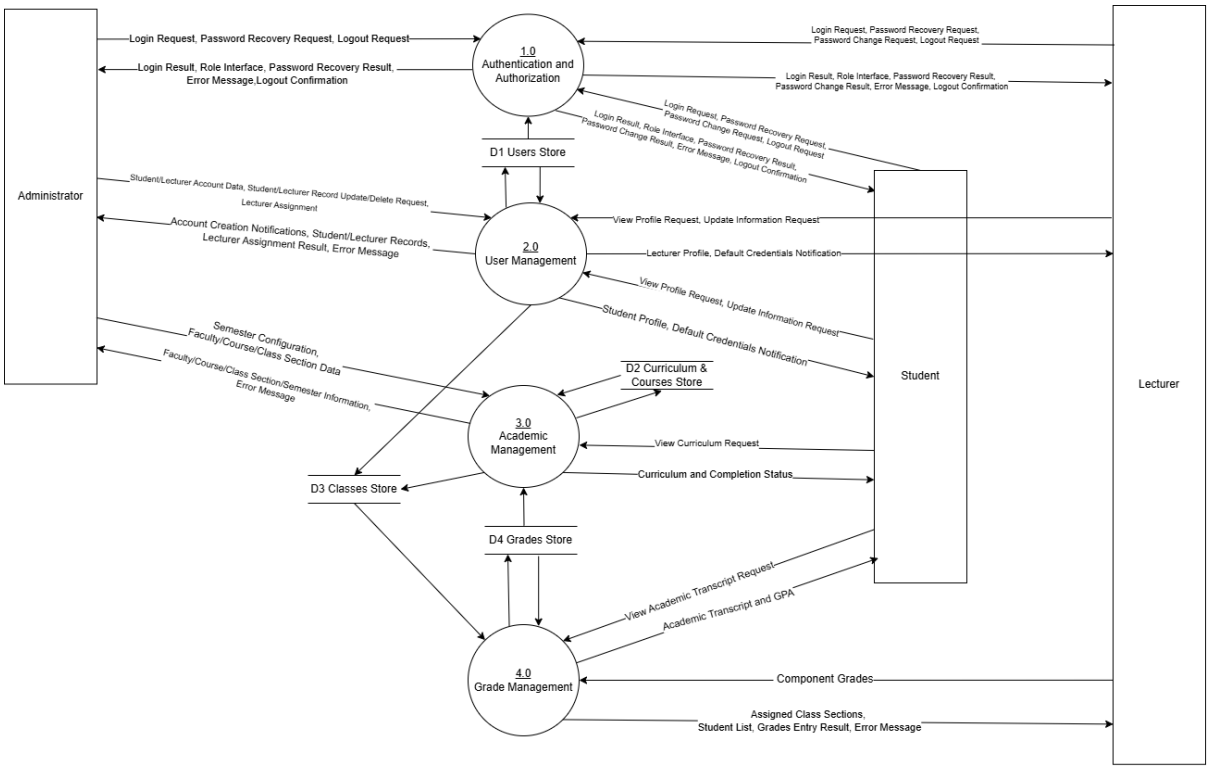


Image 2: DFD Diagram Level 1

3.2. Processes

Process ID	Process Name	Functional Description

1.0	Authentication and Authorization	Handles identity verification and role-based access control for all users (Administrator, Student, Lecturer). Manages login, password recovery, password change and logout
2.0	User Management	Manages student and lecturer accounts and records/profiles, including view, update and deletion. Handles lecturer assignment and personal contact information updates
3.0	Academic Management	Manages academic data including semester configuration, faculty, course, curriculum and class section data. Only the Administrator can configure this data
4.0	Grade Management	Manages component grade entry by Lecturers. Calculates course results and GPA for Students

3.3. Data Stores

Data Store ID	Data Store Name	Description
D1	Users Store	Stores account information of all users (username, password, role, account status), login sessions, and the detailed records/profiles of students and lecturers

D2	Curriculum and Courses Store	Stores the catalogs of faculties, majors, courses and the curriculum frameworks of each major, including compulsory and elective courses
D3	Classes Store	Stores academic semesters and the class sections opened in each semester, including schedules, rooms, maximum capacity, current enrollment, enrolled students, and lecturer assignments
D4	Grades Store	Stores component marks (midterm, assignments, attendance, final exam), final course grades with letter grades and finalized status, and students' Semester GPA and Cumulative GPA

CHAPTER 4: USE-CASE DIAGRAM

1. Use-Case Diagram

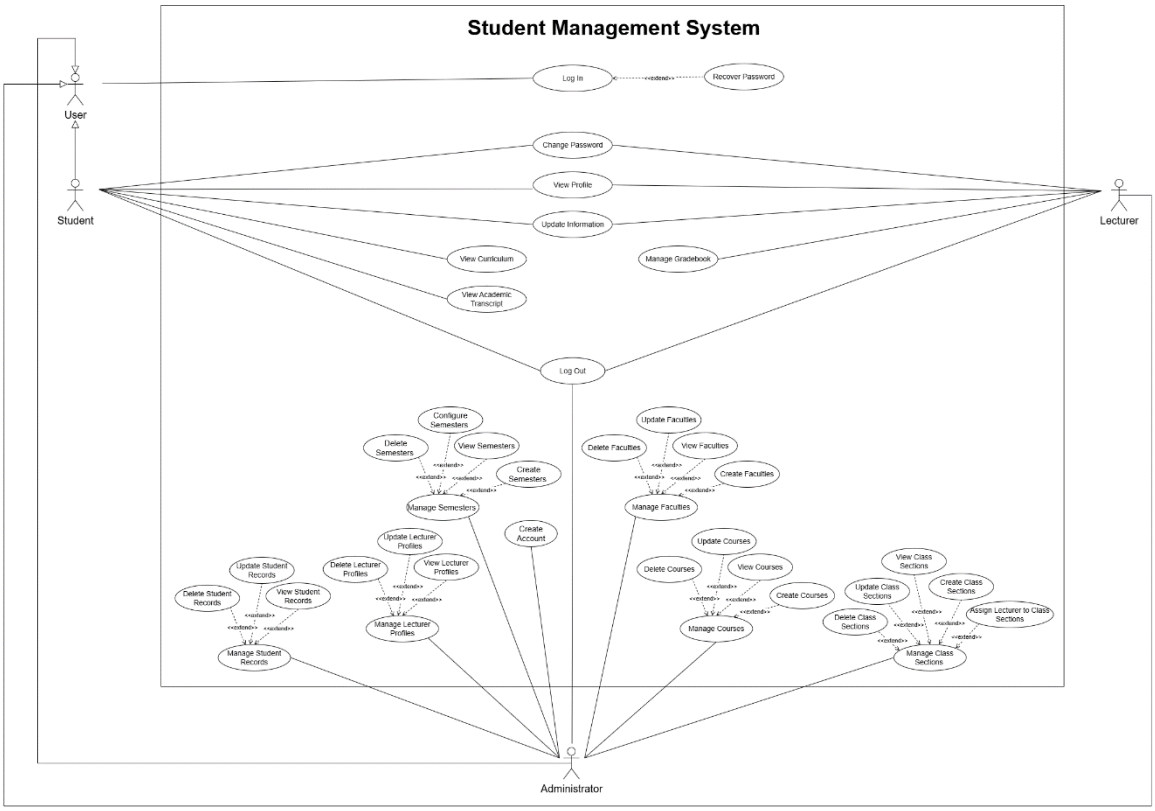


Image 3: Use-Case Diagram

1.1. List of Actor

No	Actor	Description
1	User	Represents an individual who uses the system's authentication functions, including selecting a login role (Administrator, Student or Lecturer), logging in and recovering a password. After successful login, acts as an Administrator, Student or Lecturer according to the role stored in the account

2	Student	Represents a student who can change password, view and update personal information, view the curriculum, view academic transcript and log out
3	Lecturer	Represents a lecturer who can change password, view and update personal information, manage gradebooks for assigned class sections and log out
4	Administrator	Represents the person responsible for creating accounts, managing student and lecturer records, managing academic arrangements: semesters, faculties, courses, class sections and logging out

1.2. List of Use-Case

No	Use-Case	Description
1	Log In	Allows registered users to select a login role and access the system with their credentials, then opens the functions permitted for their role
2	Recover Password	Allows users to set a new password after verifying ownership through the registered email address or phone number

3	Change Password	Allows students and lecturers to replace the current password with a new one that meets the complexity rules
4	View Profile	Allows students and lecturers to view their personal and contact information
5	Update Information	Allows students and lecturers to update their own contact information. Identification data and faculty cannot be modified
6	Log Out	Allows users to log out at any time. The system terminates the active session and clears all temporary session data
7	View Curriculum	Allows students to view the curriculum of their major and the completion status of each course
8	View Academic Transcript	Allows students to view their finalized grades, Semester GPA and Cumulative GPA
9	Manage Gradebook	Allows lecturers to record student marks. The system validates the marks and calculates the final course grade
10	Create Account	Allows the Administrator to create new student and lecturer accounts, each with its student record or lecturer profile and

		default credentials. The system validates the input and notifies the user
11	Manage Student Records	Allows the administrator to view, update and delete student records
12	View Student Records	Allows the administrator to view the list and details of student records
13	Update Student Records	Allows the administrator to modify the information and the academic or administrative status of a student record
14	Delete Student Records	Allows the administrator to delete a student record
15	Manage Lecturer Profiles	Allows the administrator to view, update, and delete lecturer profiles
16	View Lecturer Profiles	Allows the administrator to view the list and details of lecturer profiles
17	Update Lecturer Profiles	Allows the administrator to modify the information of a lecturer profile
18	Delete Lecturer Profiles	Allows the administrator to delete a lecturer profile
19	Manage Semesters	Allows the administrator to create, view, configure and delete academic semesters. All academic activities and

		class section offerings are linked to a specific semester. A semester linked to class sections or academic records cannot be deleted
20	Create Semester	Allows the administrator to create a new semester with the required fields: Semester Name, Academic Year, Start Date and End Date
21	View Semester	Allows the administrator to view the list and details of semesters
22	Configure Semester	Allows the administrator to modify the name, academic year, start date and end date of an existing semester
23	Delete Semester	Allows the administrator to delete a semester. The system rejects the deletion if the semester is linked to class sections or historical academic records
24	Manage Faculties	Allows the administrator to manage faculty records. A faculty with active courses or enrolled students cannot be deleted
25	Create Faculties	Allows the administrator to add a new faculty with Faculty Code, Faculty Name, Email and Phone Number

26	View Faculties	Allows the administrator to view the list and details of faculty records
27	Update Faculties	Allows the administrator to modify the name, email and phone number of a faculty record
28	Delete Faculties	Allows the administrator to delete a faculty record. The system rejects the deletion if the faculty currently manages active courses or has enrolled students
29	Manage Courses	Allows the administrator to manage course records. A course linked to class sections or academic records cannot be deleted
30	Create Courses	Allows the administrator to add a new course with Course Code, Course Name, Credits, Managing Faculty and Prerequisite Constraints
31	View Courses	Allows the administrator to view the list and details of courses
32	Update Courses	Allows the administrator to modify the information of a course record
33	Delete Courses	Allows the administrator to delete a course record. The system rejects the deletion if the course is linked to

		ongoing class sections or historical academic records
34	Manage Class Sections	Allows the administrator to manage the class sections of a course in a semester
35	Create Class Sections	Allows the administrator to create a class section with Class Code, Course Code, Semester, Maximum Capacity, Room and Class Schedule
36	View Class Sections	Allows the administrator to view the list and details of class sections
37	Update Class Sections	Allows the administrator to modify the information of a class section
38	Delete Class Sections	Allows the administrator to delete a class section
39	Assign Lecturer to Class Sections	Allows the administrator to assign a lecturer to a class section

2. Use-Case Log In

2.1. Summary

Allows registered users to select a login role (Administrator, Student or Lecturer) and log in to the system using a username and password

2.2 Flow of Events

2.2.1. Main Flow

1. The user opens the login page and selects the login role: Administrator, Student or Lecturer
2. The user enters the username and password, then submits the login request
3. The system validates the credentials against the Microsoft SQL Server database
4. The system identifies the user role: Administrator, Student or Lecturer and checks that it matches the selected role
5. The system displays the interface and functions permitted for that role

2.2.2. Alternative Flow

A1: If the credentials are invalid or the account is locked, the system rejects the request and displays an appropriate error message

A2: At Step 4, if the role stored in the account differs from the selected role, the system rejects the request, creates no session, and displays the same message as for invalid credentials. The system does not reveal the account's actual role

2.3. Special Requirement

The system shall respond to the login request within 15 seconds under normal operating conditions, restrict access to features and data according to the role of each user, and automatically terminate an active session after 15 minutes of inactivity, clear all session data, and redirect the user to the login page

2.4. Precondition

The user has a valid registered account created by the Administrator with an assigned role

2.5. Postconditions

Success: The user is authenticated and an active session is initiated.

Failure: No session is created and an error message is displayed.

2.6. Extension Points

Recover Password

3. Use-Case Recover Password

3.1. Summary

Allows users to recover their accounts by setting a new password after verifying account ownership through a registered email address or phone number

3.2 Flow of Events

3.2.1. Main Flow

1. The user selects the ‘’Recover Password” option on the login page
2. The user provides the registered email address or phone number and submits the recovery request
3. The system verifies the provided email address or phone number against the Microsoft SQL Server database
4. Once account ownership is verified, the system allows the user to set a new password
5. The user enters the new password and confirms it
6. The system validates the password complexity, updates the user’s record in the Microsoft SQL Server database and displays a success message

3.2.2. Alternative Flow

A1: If the email address or phone number is not registered, account ownership cannot be verified, or the new password does not meet the complexity rules, the system rejects the request and displays an error message

A2: If saving the new password to the database fails, the system rolls back all related changes, keeps the password unchanged and displays an error message

3.3. Special Requirement

The system shall respond to the password recovery request within 15 seconds under normal operating conditions, display clear error messages for invalid inputs, and confirm completion only after password updates are committed successfully to the database without exposing sensitive user credentials

3.4. Precondition

The user has a registered account in the system and cannot access their account or has requested a password reset from the Log In interface

3.5. Postconditions

Success: The new password is saved and the user can log in with it

Failure: The password remains unchanged and an error message is displayed

3.6. Extension Points

None

4. Use-Case Change Password

4.1. Summary

Allows students and lecturers to change their password by entering their current password and a new password meeting complexity rules

4.2 Flow of Events

4.2.1. Main Flow

1. The user selects the “Change Password” option
2. The system prompts the user to enter the current password, the new password, and the confirmation of the new password
3. The user enters the required information and submits the request
4. The system verifies the current password against the Microsoft SQL Server database

5. The system validates that the new password meets the password complexity rules and matches the confirmation
6. The system updates the user's record in the Microsoft SQL Server database and displays a success message

4.2.2. Alternative Flow

A1: If the current password is incorrect, the new password does not meet the complexity rules, or the new password confirmation does not match, the system rejects the request and displays an appropriate error message

A2: If saving the new password to the database fails, the system rolls back all related changes, keeps the password unchanged and displays an error messag

4.3. Special Requirement

The system shall respond to the password change request within 15 seconds under normal operating conditions, display clear error messages for invalid inputs, and confirm completion only after password updates are committed successfully to the database without exposing sensitive user credentials

4.4. Precondition

The user is authenticated as a Student or Lecturer and has an active session in the system

4.5. Postconditions

Success: The new password is saved and replaces the current password

Failure: The password remains unchanged and an error message is displayed

4.6. Extension Points

None

5. Use-Case View Profile

5.1. Summary

Allows students and lecturers to view their personal and account profile information

5.2 Flow of Events

5.2.1. Main Flow

1. The user selects the “View Profile” option
2. The system retrieves the user’s personal information from the Microsoft SQL Server database
3. The system displays the user’s profile details on the interface

5.2.2. Alternative Flow

If the personal information cannot be retrieved, the system displays an error message

5.3. Special Requirement

The system shall respond to the request within 15 seconds under normal operating conditions. The system shall restrict access so that users can only view their own profile

5.4. Precondition

The user is authenticated as a Student or Lecturer and has an active session in the system

5.5. Postconditions

Success: The personal information of the user is displayed.

Failure: An error message is displayed

5.6. Extension Points

None

6. Use-Case Update Information

6.1. Summary

Allows students and lecturers to update their own contact information while preventing modifications to identification data or affiliated faculty

6.2 Flow of Events

6.2.1. Main Flow

1. The user selects the “Update Information” option
2. The system displays the current contact information of the user
3. The user edits the contact information and submits the request
4. The system validates the submitted information
5. The system updates the user’s record in the Microsoft SQL Server database and displays a success message

6.2.2. Alternative Flow

A1: If the submitted information is invalid or the user attempts to modify identification data or affiliated faculty, the system rejects the request and displays an error message identifying the invalid field

A2: If saving the contact information to the database fails, the system rolls back all related changes, keeps the contact information unchanged and displays an error message

6.3. Special Requirement

The system shall respond to the update request within 15 seconds under normal operating conditions. The system shall not allow users to modify their identification data or affiliated faculty and if the input format is invalid, the system displays an error message identifying the invalid field

6.4. Precondition

The user is authenticated as a Student or Lecturer and has an active session in the system

6.5. Postconditions

Success: The contact information of the user is updated.

Failure: The contact information remains unchanged and an error message is displayed

6.6. Extension Points

None

7. Use-Case View Curriculum

7.1. Summary

Allows students to view the standard curriculum of their major, including compulsory courses, elective courses, credit values, and prerequisites

7.2 Flow of Events

7.2.1. Main Flow

1. The student selects the “View Curriculum” option
2. The system retrieves the standard curriculum of the student’s major and the student’s academic transcript from the Microsoft SQL Server database
3. The system marks the completion status of each course based on the academic transcript
4. The system displays the compulsory courses, elective courses, credit values, prerequisites, and completion status of each course

7.2.2. Alternative Flow

If the curriculum or the academic transcript cannot be retrieved, the system displays an error message

7.3. Special Requirement

The system shall respond to the request within 15 seconds under normal operating conditions. The system shall automatically mark and update the completion status of each course based on the student’s academic transcript and restrict access so that students can only view the curriculum of their own major

7.4. Precondition

The user is authenticated as a Student and has an active session in the system

7.5. Postconditions

Success: The curriculum and the completion status of each course are displayed

Failure: An error message is displayed

7.6. Extension Points

None

8. Use-Case View Academic Transcript

8.1. Summary

Allows a student to view their academic history and results.

8.2. Flow of Events

8.2.1. Main Flow

1. The use case starts when the student selects “Transcript”.
2. The system retrieves the student’s finalized results from the database.
3. The system displays component marks, final exam results, and letter grades per course.
4. The system displays Semester GPA and Cumulative GPA.

8.2.2. Alternative Flow

A1: At Step 2, if no finalized grades exist for the student, the system displays a message stating that no finalized grade data is available

A2: At Step 2, if database access or verification fails due to technical errors, the system displays an error message

8.3. Special Requirements

Only finalized grades are shown, unfinalized/draft marks must not be displayed.

GPA must update automatically based on finalized grades.

Response time from selecting "Transcript" to rendering the full transcript must not exceed 15 seconds under standard test load conditions

8.4. Preconditions

The user is authenticated as a Student and has an active session in the system

8.5. Postconditions

Success: The finalized grades, Semester GPA and Cumulative GPA are displayed, and data is unchanged

Failure: An error message is displayed and data is unchanged

8.6. Extension Points

None

9. Use-Case Log out

9.1. Summary

Allows a user to exit the system

9.2. Flow of Events

9.2.1. Main Flow

1. The use case starts when the user selects “Log out”.
2. The system terminates the active user session in the database
3. The system clears temporary session data.
4. Upon successful session invalidation in the database, the system redirects the user to the login page.

9.2.2. Alternative Flow

If no user activity is detected for 15 minutes, the system automatically performs Steps 2–4, invalidates the session, and redirects to the login page with a notification that the session expired due to inactivity

9.3. Special Requirements

Logout must be available at any time from any screen for logged-in users.

The login page redirection and logout success state must only occur after session deletion is confirmed in the database state.

All temporary session data and cached sensitive data must be completely cleared.

9.4. Preconditions

The user is authenticated and has an active session in the system

9.5. Postconditions

Success: Session terminated in database, temporary session data cleared, user redirected to login page.

Failure: Session remains active in the database; user context and local data remain unchanged; error message is displayed.

9.6. Extension Points

None

10. Use-Case Manage Gradebook

10.1. Summary

Allows a lecturer to record and update student grades for assigned class sections.

10.2. Flow of Events

10.2.1. Main Flow

1. The use case starts when the lecturer selects “Gradebook”.
2. The lecturer selects an assigned section.
3. The system verifies assignments and displays the list of students along with grade components (midterm, assignments, attendance, final exam).
4. The lecturer enters or updates component marks and clicks “Save”.
5. The system validates the entered marks
6. The system calculates the final grade using the course’s weighted grading formula, determines the letter grade, updates GPA, and saves all data to the database.
7. Upon successful database update, the system displays a success message.

10.2.2. Alternative Flow

A1: At Step 5, if any entered mark is outside the allowed range (0.0 – 10.0) or invalid, the system rejects the input and displays an error message identifying the invalid field

A2: At Step 6, if saving grade data to the database fails due to a system error or failed verification, the system rolls back the transaction, maintains original grades, and displays an error message

10.3. Special Requirements

Final numerical grades, letter grades, and GPA impact must be calculated automatically; students can only view grades after they are finalized.

Response time for grade validation, formula calculation, and database persistence must not exceed 15 seconds under standard test load conditions

10.4. Preconditions

The lecturer has an active session and is assigned to the class section. The section includes enrolled students, and the course has a defined grading formula

10.5. Postconditions

Success: Component marks, final grade, letter grade, and GPA are saved in the database.

Failure: Grade records remain unchanged in the database.

10.6. Extension Points

None

11. Use-Case Create Account

11.1. Summary

Allows the Administrator to create new student and lecturer accounts, each with its student record or lecturer profile and default credentials

11.2. Flow of Events

11.2.1. Main Flow

1. The use case starts when the Administrator selects “Create account”.
2. The Administrator selects the account type: student/lecturer
3. The Administrator enters the required personal, identification, contact, and faculty or major information according to the selected account type
4. The system validates the submitted information, including mandatory fields and unique-field constraints
5. The system automatically generates the appropriate Student ID or Lecturer ID and creates the student record or lecturer profile, together with the account with the initial credentials
6. Upon successful database commitment, the system notifies the Administrator with a success message and sends account details to the user.

11.2.2. Alternative Flow

A1: At Step 4, if mandatory fields are missing or unique constraints are violated, the system rejects the request and displays an error message naming the invalid field

A2: At Step 5, if saving the new record or profile and the account to the database fails due to technical errors or system constraints, the system rolls back the transaction, creates no record, profile or account, and displays an error message

11.3. Special Requirements

Only Administrators are authorized to create accounts, based on official admission or personnel records.

Response time for validating input, generating IDs, saving to database, and returning the success message must not exceed 15 seconds under standard test load conditions

11.4. Preconditions

The user is authenticated as an Administrator and has an active session in the system

11.5. Postconditions

Success: New account, ID, and record or profile are stored in the database, and notification is sent.

Failure: No account, record or profile is created; database remains unchanged.

11.6. Extension Points

None

12. Use-Case Manage Student Records

12.1. Summary

Allows the Administrator to view, update, and delete student records, while students can view and update their own contact information

12.2. Flow of Events

12.2.1. Main Flow

1. The use case starts when the Administrator selects “Student management”.
2. The system displays the student list
3. The Administrator chooses to view, update, or delete
4. For View: The system displays the details of the selected record
5. For Update: The Administrator enter profile data and academic/administrative status; the system validates and saves.
6. For Delete: The system checks whether the student record can be deleted without violating data integrity constraints. If deletion is allowed, the system asks the Administrator for confirmation. After confirmation, the system deletes the record and displays the result.
7. Upon successful database commitment, the system displays a success message with the operation result.

12.2.2. Alternative Flow

A1: Student self-update:

1. The student opens their own profile.
2. The system displays it; only contact fields: email, phone, … are editable.
3. The student edits contact details and clicks Save.
4. If the student tries to modify identification data or affiliated faculty, the system rejects the update and displays an error message stating that students may only modify contact information

A2: At Step 4, if a mandatory field is missing or a unique field is duplicated, the system rejects the request and displays an error message identifying the invalid field

A3: At Step 5, if deletion violates data integrity constraints, the system rejects the request and displays the reason

A4: At Step 5, if the Administrator cancels the confirmation, no record is deleted

A5: At Step 4 or 5, if saving or deleting fails due to technical errors or system constraints, the system rolls back all related changes, keeps the student record unchanged, and displays an error message

12.3. Special Requirements

Each student record must explicitly include an academic or administrative status. Students can only access their own records and modify contact details, identification data and affiliated faculty must be read-only for students.

Response time for validating, saving, or deleting records in the database must not exceed 15 seconds under standard test load conditions

12.4. Preconditions

The user is authenticated as an Administrator and has an active session in the system

12.5. Postconditions

Success: Student record is updated, or deleted in the database.

Failure: Student database records remain unchanged; error message is displayed.

12.6. Extension Points

View Student Records

Update Student Records

Delete Student Records

13. Use-Case View Student Records

13.1. Summary

Allows the Administrator to view the list and details of student records, including each student's academic or administrative status

13.2. Flow of Events

13.2.1. Main Flow

1. The use case starts when the Administrator selects "Student management" and chooses "View"
2. The system retrieves the student records from the Microsoft SQL Server database and displays the student list
3. The Administrator selects a student record from the list
4. The system displays the details of the selected record: Student ID, Citizen ID Number, Full Name, Date of Birth, Email, Gender, Phone Number, Faculty/Major, and academic or administrative status

13.2.2. Alternative Flow

A1: At Step 2, if no student record exists or no record matches the search criteria, the system displays a message stating that no student records were found

A2: At Steps 2 and 4, if the student records cannot be retrieved from the database, the system displays an error message and no data is displayed

13.3. Special Requirements

Only the Administrator is authorized to view the records of all students. Each displayed student record must include an academic or administrative status

Response time for retrieving and displaying student records must not exceed 15 seconds under standard test load conditions

13.4. Preconditions

The user is authenticated as an Administrator and has an active session in the system

13.5. Postconditions

Success: The student list and the details of the selected student record are displayed
Failure: An error message is displayed; student database records remain unchanged

13.6. Extension Points

None

14. Use-Case Update Student Records

14.1. Summary

Allows the Administrator to modify the information and the academic or administrative status of a student record

14.2. Flow of Events

14.2.1. Main Flow

1. The use case starts when the Administrator selects “Student management” and chooses “Update”
2. The system displays the student list
3. The Administrator selects a student record from the list
4. The system retrieves the selected record from the Microsoft SQL Server database and displays its current information in an editable form
5. The Administrator modifies the student information and the academic or administrative status, then submits the form
6. The system validates all inputs, including mandatory fields and unique-field constraints
7. The system saves the updated student record to the database

8. Upon successful database commitment, the system displays a success message with the operation result

14.2.2. Alternative Flow

A1: At Step 2, if no student record exists or no record matches the search criteria, the system displays a message stating that no student records were found

A2: At Step 4, if the selected record cannot be retrieved from the database, the system displays an error message and no data is displayed

A3: At Step 6, if a mandatory field is missing or a unique field is duplicated, the system rejects the request and displays an error message identifying the invalid field

A4: At Step 7, if saving fails due to technical errors or system constraints, the system rolls back all related changes, keeps the student record unchanged, and displays an error message

14.3. Special Requirements

Only the Administrator is authorized to update student records. Each student record must explicitly include an academic or administrative status, and the Student ID is generated automatically and cannot be modified

Response time for validating and saving the record in the database must not exceed 15 seconds under standard test load conditions

14.4. Preconditions

The user is authenticated as an Administrator and has an active session in the system

14.5. Postconditions

Success: The student record is updated in the database and a success message is displayed

Failure: The student record remains unchanged and an error message is displayed

14.6. Extension Points

None

15. Use-Case Delete Student Records

15.1. Summary

Allows the Administrator to delete a student record from the system

15.2. Flow of Events

15.2.1. Main Flow

1. The use case starts when the Administrator selects “Student management” and chooses “Delete”
2. The system displays the student list
3. The Administrator selects a student record from the list
4. The system checks whether the student record can be deleted without violating data integrity constraints
5. If deletion is allowed, the system asks the Administrator for confirmation
6. After confirmation, the system deletes the student record from the Microsoft SQL Server database
7. Upon successful database commitment, the system displays a success message with the operation result

15.2.2. Alternative Flow

A1: At Step 2, if no student record exists or no record matches the search criteria, the system displays a message stating that no student records were found

A2: At Step 4, if deletion violates data integrity constraints, the system rejects the request, keeps the student record unchanged, and displays the reason

A3: At Step 5, if the Administrator cancels the confirmation, no record is deleted

A4: At Step 6, if deleting fails due to technical errors or system constraints, the system rolls back all related changes, keeps the student record unchanged, and displays an error message

15.3. Special Requirements

Only the Administrator is authorized to delete student records. A student record that violates data integrity constraints cannot be deleted.

Response time for checking, deleting, and confirming the operation in the database must not exceed 15 seconds under standard test load conditions

15.4. Preconditions

The user is authenticated as an Administrator and has an active session in the system

15.5. Postconditions

Success: The student record is deleted from the database and a success message is displayed

Failure: The student record remains unchanged and an error message is displayed

15.6. Extension Points

None

16. Use-Case Manage Lecturer Profiles

16.1. Summary

Allows the Administrator to view, update, and delete lecturer profiles, while lecturers can view and update their own contact information

16.2. Flow of Events

16.2.1. Main Flow

1. The use case starts when the Administrator selects “Lecturer management”
2. The system displays the lecturer list
3. The Administrator selects a lecturer profile and chooses to view, update, or delete
4. For View: The system displays the details of the selected lecturer profile
5. For Update: The Administrator enter lecturer profile data, the system validates and saves

6. For Delete: The system checks whether the lecturer profile can be deleted without violating data integrity constraints. If deletion is allowed, the system asks the Administrator for confirmation. After confirmation, the system deletes the profile and displays the result
7. Upon successful database commitment, the system displays a success message

16.2.2. Alternative Flow

A1: Lecturer self-update:

1. The lecturer opens their own profile
2. The system displays it; only contact fields are editable
3. The lecturer edits contact details and clicks Save
4. The system validates and updates the database
5. If the lecturer attempts to modify identification data or affiliated faculty, the system rejects the update, performs rollback, and displays an error message stating that lecturers may only modify contact information

A2: At Step 5, if required fields are missing or unique constraints are violated, the system rejects the request, performs rollback, and displays an error message naming the invalid field

A3: At Step 6, if confirmation is cancelled by the Administrator, the system cancels the operation and leaves database data unchanged

A4: At Step 6, if deletion violates data integrity constraints, the system rejects the request, keeps the lecturer profile unchanged, and displays the reason

A5: At Step 5 or 6, if saving or deleting fails due to technical errors or system constraints, the system rolls back all related changes, keeps the lecturer profile unchanged, and displays an error message

16.3. Special Requirements

Lecturers may only access their own profiles and update contact details. Deletion must strictly preserve data integrity

Response time for updating, or deleting lecturer profiles in the database must not exceed 15 seconds under standard test load conditions

16.4. Preconditions

The user is authenticated as an Administrator and has an active session in the system

16.5. Postconditions

Success: Lecturer profile is updated, or deleted in the database.

Failure: Lecturer records remain unchanged in the database; error message is displayed.

16.6. Extension Points

View Lecturer Profiles

Update Lecturer Profiles

Delete Lecturer Profiles

17. Use-Case View Lecturer Profiles

17.1. Summary

Allows the Administrator to view the list and details of lecturer profiles

17.2. Flow of Events

17.2.1. Main Flow

1. The use case starts when the Administrator selects “Lecturer management” and chooses “View”
2. The system retrieves the lecturer profiles from the Microsoft SQL Server database and displays the lecturer list
3. The Administrator selects a lecturer profile from the list
4. The system displays the details of the selected profile: Lecturer ID, Citizen ID Number, Full Name, Date of Birth, Email, Gender, Phone Number, and Faculty

17.2.2. Alternative Flow

A1: At Step 2, if no lecturer profile exists or no profile matches the search criteria, the system displays a message stating that no lecturer profiles were found

A2: At Steps 2 and 4, if the lecturer profiles cannot be retrieved from the database, the system displays an error message and no data is displayed

17.3. Special Requirements

Only the Administrator is authorized to view the profiles of all lecturers.
Response time for retrieving and displaying lecturer profiles must not exceed 15 seconds under standard test load conditions

17.4. Preconditions

The user is authenticated as an Administrator and has an active session in the system

17.5. Postconditions

Success: The lecturer list and the details of the selected lecturer profile are displayed
Failure: An error message is displayed; lecturer database records remain unchanged

17.6. Extension Points

None

18. Use-Case Update Lecturer Profiles

18.1. Summary

Allows the Administrator to modify the information of a lecturer profile

18.2. Flow of Events

18.2.1. Main Flow

1. The use case starts when the Administrator selects “Lecturer management” and chooses “Update”
2. The system displays the lecturer list
3. The Administrator selects a lecturer profile from the list

4. The system retrieves the selected profile from the Microsoft SQL Server database and displays its current information in an editable form
5. The Administrator modifies the lecturer information, then submits the form
6. The system validates all inputs, including mandatory fields and unique-field constraints
7. The system saves the updated lecturer profile to the database
8. Upon successful database commitment, the system displays a success message with the operation result

18.2.2. Alternative Flow

A1: At Step 2, if no lecturer profile exists or no profile matches the search criteria, the system displays a message stating that no lecturer profiles were found

A2: At Step 4, if the selected profile cannot be retrieved from the database, the system displays an error message and no data is displayed

A3: At Step 6, if a mandatory field is missing or a unique field is duplicated, the system rejects the request and displays an error message identifying the invalid field

A4: At Step 7, if saving fails due to technical errors or system constraints, the system rolls back all related changes, keeps the lecturer profile unchanged, and displays an error message

18.3. Special Requirements

Only the Administrator is authorized to update lecturer profiles. The Lecturer ID is generated automatically and cannot be modified

Response time for validating and saving the profile in the database must not exceed 15 seconds under standard test load conditions

18.4. Preconditions

The user is authenticated as an Administrator and has an active session in the system

18.5. Postconditions

Success: The lecturer profile is updated in the database and a success message is displayed

Failure: The lecturer profile remains unchanged and an error message is displayed

18.6. Extension Points

None

19. Use-Case Delete Lecturer Profiles

19.1. Summary

Allows the Administrator to delete a lecturer profile from the system

19.2. Flow of Events

19.2.1. Main Flow

1. The use case starts when the Administrator selects “Lecturer management” and chooses “Delete”
2. The system displays the lecturer list
3. The Administrator selects a lecturer profile from the list
4. The system checks whether the lecturer profile can be deleted without violating data integrity constraints
5. If deletion is allowed, the system asks the Administrator for confirmation
6. After confirmation, the system deletes the lecturer profile from the Microsoft SQL Server database
7. Upon successful database commitment, the system displays a success message with the operation result

19.2.2. Alternative Flow

A1: At Step 2, if no lecturer profile exists or no profile matches the search criteria, the system displays a message stating that no lecturer profiles were found

A2: At Step 4, if deletion violates data integrity constraints, the system rejects the request, keeps the lecturer profile unchanged, and displays the reason

A3: At Step 5, if the Administrator cancels the confirmation, no profile is deleted

A4: At Step 6, if deleting fails due to technical errors or system constraints, the system rolls back all related changes, keeps the lecturer profile unchanged, and displays an error message

19.3. Special Requirements

Only the Administrator is authorized to delete lecturer profiles. A lecturer profile that violates data integrity constraints cannot be deleted

Response time for checking, deleting, and confirming the operation in the database must not exceed 15 seconds under standard test load conditions

19.4. Preconditions

The user is authenticated as an Administrator and has an active session in the system

19.5. Postconditions

Success: The lecturer profile is deleted from the database and a success message is displayed

Failure: The lecturer profile remains unchanged and an error message is displayed

19.6. Extension Points

None

20. Use-Case Manage Semesters

20.1. Summary

Allows the Administrator to create, view, configure, and delete academic semesters. All academic activities and class section offerings are linked to a specific semester.

20.2. Flow of Events

20.2.1. Main Flow

1. The use case starts when the Administrator selects “Semester management”.

2. The system displays the semester list.
3. The Administrator chooses to create, view, configure, or delete
4. For View: The system displays the details of the selected semester: Semester Name, Academic Year, Start Date and End Date
5. For Create/Configure: The Administrator enters Semester Name, Academic Year, Start Date and End Date; the system validates mandatory fields, date validity and unique-field constraints, then saves to the database
6. For Delete: The system checks whether the semester is linked to class sections or historical academic records. If deletion is allowed, the system asks the Administrator for confirmation. After confirmation, the system deletes the semester and displays the result
7. Upon successful database commitment, the system displays a success message.

20.2.2. Alternative Flow

A1: At Step 2, if no semester exists or no semester matches the search criteria, the system displays a message stating that no semesters were found

A2: At Step 5, if a mandatory field is missing, a date is invalid, or the semester already exists, the system rejects the request, performs rollback, and displays an error message identifying the invalid field

A3: At Step 6, if the semester is linked to class sections or historical academic records, the system rejects the deletion, keeps the semester unchanged, and displays the reason

A4: At Step 6, if the Administrator cancels the confirmation, the system cancels the operation and leaves database data unchanged

A5: At Step 5 or 6, if saving or deleting fails due to technical errors or system constraints, the system rolls back all related changes, keeps the semester unchanged, and displays an error message

20.3. Special Requirements

Only the Administrator is authorized to manage semesters, and all academic activities and class section offerings must be linked to a specific semester. A semester that is linked to class sections or historical academic records cannot be deleted

Response time for validating timelines and saving semester data to the database must not exceed 15 seconds under standard test load conditions

20.4. Preconditions

The user is authenticated as an Administrator and has an active session in the system

20.5. Postconditions

Success: Semester data is created, configured, or deleted in the database, or the semester details are displayed

Failure: Semester database records remain unchanged; error message is displayed.

20.6. Extension Points

Create Semesters

View Semesters

Configure Semesters

Delete Semesters

21. Use-Case Create Semesters

21.1. Summary

Allows the Administrator to create a new academic semester

21.2. Flow of Events

21.2.1. Main Flow

1. The use case starts when the Administrator selects “Semester management” and chooses “Create”
2. The system displays the semester creation form

3. The Administrator enters Semester Name, Academic Year, Start Date and End Date, then submits the form
4. The system validates all inputs, including mandatory fields, date validity (the End Date must be after the Start Date) and unique-field constraints
5. The system saves the new semester to the Microsoft SQL Server database
6. Upon successful database commitment, the system displays a success message with the operation result

21.2.2. Alternative Flow

A1: At Step 4, if a mandatory field is missing, a date is invalid, or the semester already exists, the system rejects the request and displays an error message identifying the invalid field

A2: At Step 5, if saving fails due to technical errors or system constraints, the system rolls back all related changes, creates no semester, and displays an error message

21.3. Special Requirements

Only the Administrator is authorized to create semesters. Each semester must include Semester Name, Academic Year, Start Date and End Date, and all academic activities and class section offerings are linked to a specific semester

Response time for validating and saving the semester in the database must not exceed 15 seconds under standard test load conditions

21.4. Preconditions

The user is authenticated as an Administrator and has an active session in the system

21.5. Postconditions

Success: The new semester is stored in the database and a success message is displayed

Failure: No semester is created, the database remains unchanged, and an error message is displayed

21.6. Extension Points

None

22. Use-Case View Semesters

22.1. Summary

Allows the Administrator to view the list and details of academic semesters

22.2. Flow of Events

22.2.1. Main Flow

1. The use case starts when the Administrator selects “Semester management” and chooses “View”
2. The system retrieves the semesters from the Microsoft SQL Server database and displays the semester list
3. The Administrator selects a semester from the list
4. The system displays the details of the selected semester: Semester Name, Academic Year, Start Date and End Date

22.2.2. Alternative Flow

A1: At Step 2, if no semester exists or no semester matches the search criteria, the system displays a message stating that no semesters were found

A2: At Steps 2 and 4, if the semesters cannot be retrieved from the database, the system displays an error message and no data is displayed

22.3. Special Requirements

Only the Administrator is authorized to view semesters
Response time for retrieving and displaying semesters must not exceed 15 seconds under standard test load conditions

22.4. Preconditions

The user is authenticated as an Administrator and has an active session in the system

22.5. Postconditions

Success: The semester list and the details of the selected semester are displayed
Failure: An error message is displayed; semester database records remain unchanged

22.6. Extension Points

None

23. Use-Case Configure Semesters

23.1. Summary

Allows the Administrator to modify the Semester Name, Academic Year, Start Date and End Date of an existing academic semester

23.2. Flow of Events

23.2.1. Main Flow

1. The use case starts when the Administrator selects “Semester management” and chooses “Configure”
2. The system displays the semester list
3. The Administrator selects a semester from the list
4. The system retrieves the selected semester from the Microsoft SQL Server database and displays its current information in an editable form
5. The Administrator modifies the semester information, then submits the form
6. The system validates all inputs, including mandatory fields, date validity (the End Date must be after the Start Date) and unique-field constraints
7. The system saves the updated semester to the database
8. Upon successful database commitment, the system displays a success message with the operation result

23.2.2. Alternative Flow

A1: At Step 2, if no semester exists or no semester matches the search criteria, the system displays a message stating that no semesters were found

A2: At Step 4, if the selected semester cannot be retrieved from the database, the system displays an error message and no data is displayed

A3: At Step 6, if a mandatory field is missing, a date is invalid, or the semester already exists, the system rejects the request and displays an error message identifying the invalid field

A4: At Step 7, if saving fails due to technical errors or system constraints, the system rolls back all related changes, keeps the semester unchanged, and displays an error message

23.3. Special Requirements

Only the Administrator is authorized to configure semesters. All academic activities and class section offerings remain linked to the configured semester
Response time for validating and saving the semester in the database must not exceed 15 seconds under standard test load conditions

23.4. Preconditions

The user is authenticated as an Administrator and has an active session in the system

23.5. Postconditions

Success: The semester is updated in the database and a success message is displayed
Failure: The semester remains unchanged and an error message is displayed

23.6. Extension Points

None

24. Use-Case Delete Semesters

24.1. Summary

Allows the Administrator to delete an academic semester from the system

24.2. Flow of Events

24.2.1. Main Flow

1. The use case starts when the Administrator selects “Semester management” and chooses “Delete”
2. The system displays the semester list

3. The Administrator selects a semester from the list
4. The system checks whether the semester is linked to class sections or historical academic records
5. If deletion is allowed, the system asks the Administrator for confirmation
6. After confirmation, the system deletes the semester from the Microsoft SQL Server database
7. Upon successful database commitment, the system displays a success message with the operation result

24.2.2. Alternative Flow

A1: At Step 2, if no semester exists or no semester matches the search criteria, the system displays a message stating that no semesters were found

A2: At Step 4, if the semester is linked to class sections or historical academic records, the system rejects the request, keeps the semester unchanged, and displays the reason

A3: At Step 5, if the Administrator cancels the confirmation, no semester is deleted

A4: At Step 6, if deleting fails due to technical errors or system constraints, the system rolls back all related changes, keeps the semester unchanged, and displays an error message

24.3. Special Requirements

Only the Administrator is authorized to delete semesters. A semester that is linked to class sections or historical academic records cannot be deleted
Response time for checking, deleting, and confirming the operation in the database must not exceed 15 seconds under standard test load conditions

24.4. Preconditions

The user is authenticated as an Administrator and has an active session in the system

24.5. Postconditions

Success: The semester is deleted from the database and a success message is displayed

Failure: The semester remains unchanged and an error message is displayed

24.6. Extension Points

None

25. Use-Case Manage Faculties

25.1. Summary

Allows the Administrator to create, view, update, and delete faculties.

25.2. Flow of Events

25.2.1. Main Flow

1. The use case starts when the Administrator selects “Faculty management”.
2. The system queries the database and displays the faculty list.
3. The Administrator chooses to create, view, update, or delete.
4. For View: The system displays the details of the selected faculty: Faculty Code, Name, Email, Phone
5. For Create/Update: The Administrator enters Faculty Code, Name, Email, Phone. The system validates unique Faculty Code and required fields, then saves to the database.
6. For Delete: The system checks database constraints to verify if the faculty manages active courses or has enrolled students. If clear, the system prompts for confirmation. Upon confirmation, the system deletes the faculty from the database.
7. Upon successful database commitment, the system displays a success message with the result.

25.2.2. Alternative Flow

A1: At Step 4, if required fields are missing or the Faculty Code already exists in the database, the system rejects the request, performs rollback, and displays an error message identifying the missing mandatory data or the duplicate Faculty Code

A2: At Step 5, if the faculty manages active courses or has enrolled students linked in the database, the system rejects deletion, performs rollback, and displays an error message stating that the faculty manages active courses or has enrolled students

A3: At Step 5, if the Administrator cancels the confirmation dialog, the system cancels the operation and leaves database data unchanged

25.3. Special Requirements

Faculty Code must be unique. Deletion must strictly be rejected if the faculty manages active courses or has enrolled students.

Response time for creating, updating, or deleting faculty data in the database must not exceed 15 seconds under standard test load conditions

25.4. Preconditions

The user is authenticated as an Administrator and has an active session in the system

25.5. Postconditions

Success: Faculty data is created, updated, or deleted in the database.

Failure: Faculty database records remain unchanged; error message is displayed.

25.6. Extension Points

Create Faculties

View Faculties

Update Faculties

Delete Faculties

26. Use-Case Create Faculties

26.1. Summary

Allows the Administrator to create a new faculty record with Faculty Code, Faculty Name, Email and Phone Number

26.2. Flow of Events

26.2.1. Main Flow

1. The use case starts when the Administrator selects “Faculty management” and chooses “Create”
2. The system displays the faculty creation form
3. The Administrator enters Faculty Code, Faculty Name, Email and Phone Number, then submits the form
4. The system validates all inputs, including mandatory fields, input formats and unique-field constraints
5. The system saves the new faculty record to the Microsoft SQL Server database
6. Upon successful database commitment, the system displays a success message with the operation result

26.2.2. Alternative Flow

A1: At Step 4, if a mandatory field is missing, an input is invalid, or the faculty already exists, the system rejects the request and displays an error message identifying the invalid field

A2: At Step 5, if saving fails due to technical errors or system constraints, the system rolls back all related changes, creates no faculty record, and displays an error message

26.3. Special Requirements

Only the Administrator is authorized to create faculty records. Each faculty record must include Faculty Code, Faculty Name, Email and Phone Number
Response time for validating and saving the faculty record in the database must not exceed 15 seconds under standard test load conditions

26.4. Preconditions

The user is authenticated as an Administrator and has an active session in the system

26.5. Postconditions

Success: The new faculty record is stored in the database and a success message is displayed

Failure: No faculty record is created, the database remains unchanged, and an error message is displayed

26.6. Extension Points

None

27. Use-Case View Faculties

27.1. Summary

Allows the Administrator to view the list and details of faculty records

27.2. Flow of Events

27.2.1. Main Flow

1. The use case starts when the Administrator selects “Faculty management” and chooses “View”
2. The system retrieves the faculty records from the Microsoft SQL Server database and displays the faculty list
3. The Administrator selects a faculty from the list
4. The system displays the details of the selected faculty: Faculty Code, Faculty Name, Email and Phone Number

27.2.2. Alternative Flow

A1: At Step 2, if no faculty record exists or no record matches the search criteria, the system displays a message stating that no faculties were found

A2: At Steps 2 and 4, if the faculty records cannot be retrieved from the database, the system displays an error message and no data is displayed

27.3. Special Requirements

Only the Administrator is authorized to view faculty records

Response time for retrieving and displaying faculty records must not exceed 15 seconds under standard test load conditions

27.4. Preconditions

The user is authenticated as an Administrator and has an active session in the system

27.5. Postconditions

Success: The faculty list and the details of the selected faculty are displayed

Failure: An error message is displayed; faculty database records remain unchanged

27.6. Extension Points

None

28. Use-Case Update Faculties

28.1. Summary

Allows the Administrator to modify the Faculty Name, Email and Phone Number of a faculty record

28.2. Flow of Events

28.2.1. Main Flow

1. The use case starts when the Administrator selects “Faculty management” and chooses “Update”
2. The system displays the faculty list
3. The Administrator selects a faculty from the list
4. The system retrieves the selected faculty record from the Microsoft SQL Server database and displays its current information in an editable form
5. The Administrator modifies the faculty information, then submits the form
6. The system validates all inputs, including mandatory fields, input formats and unique-field constraints

7. The system saves the updated faculty record to the database
8. Upon successful database commitment, the system displays a success message with the operation result

28.2.2. Alternative Flow

A1: At Step 2, if no faculty record exists or no record matches the search criteria, the system displays a message stating that no faculties were found

A2: At Step 4, if the selected faculty record cannot be retrieved from the database, the system displays an error message and no data is displayed

A3: At Step 6, if a mandatory field is missing, an input is invalid, or a unique field is duplicated, the system rejects the request and displays an error message identifying the invalid field

A4: At Step 7, if saving fails due to technical errors or system constraints, the system rolls back all related changes, keeps the faculty record unchanged, and displays an error message

28.3. Special Requirements

Only the Administrator is authorized to update faculty records. The Faculty Code cannot be modified

Response time for validating and saving the faculty record in the database must not exceed 15 seconds under standard test load conditions

28.4. Preconditions

The user is authenticated as an Administrator and has an active session in the system

28.5. Postconditions

Success: The faculty record is updated in the database and a success message is displayed

Failure: The faculty record remains unchanged and an error message is displayed

28.6. Extension Points

None

29. Use-Case Delete Faculties

29.1. Summary

Allows the Administrator to delete a faculty record from the system

29.2. Flow of Events

29.2.1. Main Flow

1. The use case starts when the Administrator selects “Faculty management” and chooses “Delete”
2. The system displays the faculty list
3. The Administrator selects a faculty from the list
4. The system checks whether the faculty currently manages active courses or has enrolled students
5. If deletion is allowed, the system asks the Administrator for confirmation
6. After confirmation, the system deletes the faculty record from the Microsoft SQL Server database
7. Upon successful database commitment, the system displays a success message with the operation result

29.2.2. Alternative Flow

A1: At Step 2, if no faculty record exists or no record matches the search criteria, the system displays a message stating that no faculties were found

A2: At Step 4, if the faculty currently manages active courses or has enrolled students, the system rejects the request, keeps the faculty record unchanged, and displays the reason

A3: At Step 5, if the Administrator cancels the confirmation, no faculty record is deleted

A4: At Step 6, if deleting fails due to technical errors or system constraints, the system rolls back all related changes, keeps the faculty record unchanged, and displays an error message

29.3. Special Requirements

Only the Administrator is authorized to delete faculty records. A faculty that currently manages active courses or has enrolled students cannot be deleted

Response time for checking, deleting, and confirming the operation in the database must not exceed 15 seconds under standard test load conditions

29.4. Preconditions

The user is authenticated as an Administrator and has an active session in the system

29.5. Postconditions

Success: The faculty record is deleted from the database and a success message is displayed

Failure: The faculty record remains unchanged and an error message is displayed

29.6. Extension Points

None

30. Use-Case Manage Courses

30.1. Summary

Allows the Administrator to create, view, update, and delete courses.

30.2. Flow of Events

30.2.1. Main Flow

1. The use case starts when the Administrator selects “Course management”.
2. The system displays the course list.
3. The Administrator chooses to create, view, update, or delete.
4. For View: the system displays the details of the selected course: Course Code, Course Name, Credits, Managing Faculty and Prerequisite Constraints

5. For Create/Update: The Administrator enters Course Code, Course Name, Credits, Managing Faculty and Prerequisite Constraints. The system validates mandatory fields, the unique Course Code and the existing Managing Faculty, then saves to the database
6. For Delete: The system checks database links to verify if the course is linked to ongoing class sections or historical academic records. If clear, the system prompts for confirmation. Upon confirmation, the system deletes the course from the database.
7. Upon successful database commitment, the system displays a success message.

30.2.2. Alternative Flow

A1: At Step 2, if no course exists or no course matches the search criteria, the system displays a message stating that no courses were found

A2: At Step 5, if a mandatory field is missing, an input is invalid, the Course Code already exists, or the Managing Faculty does not exist, the system rejects the request, performs rollback, and displays an error message identifying the invalid field

A3: At Step 6, if the course is linked to ongoing class sections or historical academic records, the system rejects the deletion, keeps the course unchanged, and displays the reason

A4: At Step 6, if the Administrator cancels the confirmation, the system cancels the operation and leaves database data unchanged

A5: At Step 5 or 6, if saving or deleting fails due to technical errors or system constraints, the system rolls back all related changes, keeps the course unchanged, and displays an error message

30.3. Special Requirements

Only the Administrator is authorized to manage courses. Each course must include Course Code, Course Name, Credits, Managing Faculty and Prerequisite Constraints. The Course Code must be unique and cannot be modified. A course that is linked to ongoing class sections or historical academic records cannot be deleted.

Response time for validating and persisting course updates or deletions in the database must not exceed 15 seconds under standard test load conditions

30.4. Preconditions

The user is authenticated as an Administrator and has an active session in the system; for create/update, the Managing Faculty exists; for view/update/delete, the course exists

30.5. Postconditions

Success: Course data is created, updated, or deleted in the database, or the course details are displayed

Failure: Course database records remain unchanged; error message is displayed.

30.6. Extension Points

Create Courses

View Courses

Update Courses

Delete Courses

31. Use-Case Create Courses

31.1. Summary

Allows the Administrator to create a new course by entering its information, such as Course Code, Course Name, Credits Value, Prerequisite Constraints and Affiliated Faculty

31.2. Flow of Events

31.2.1. Main Flow

1. The Administrator selects the “Manage Courses” option and chooses the “Create Course” function
2. The system displays the course creation form

3. The Administrator enters the course information (course code, course name, credit value, prerequisites constraints, affiliated faculty) and submits the request
4. The system validates the submitted information, including required fields, valid formats, that the course code is unique, that the credit value is a positive number, and that the selected prerequisites and faculty exist
5. The system creates the new course record in the Microsoft SQL Server database
6. The system displays a success message and shows the newly created course in the course list

31.2.2. Alternative Flow

A1: At Step 4, if the submitted information is invalid or incomplete, the system rejects the request, does not create the course and displays an error message identifying the invalid field

A2: At Step 4, if the course code already exists, the system rejects the request and displays an error message stating that the course already exists

A3: At Step 5, if the new record cannot be committed to the database, the system creates no course and displays an error message

31.3. Special Requirements

The system shall respond to the create request within 15 seconds under normal operating conditions, restrict this function to the Administrator only, display clear error messages for invalid inputs, and confirm completion only after the new course is committed successfully to the database

31.4. Preconditions

The user is authenticated as an Administrator and has an active session in the system, and the affiliated faculty of the course already exists in the system

31.5. Postconditions

Success: The new course is created and appears in the course list

Failure: No course is created and an error message is displayed

31.6. Extension Points

None

32. Use-Case View Courses

32.1. Summary

Allows the Administrator to view the list of courses and the detailed information of a selected course, including course code, course name, credit value, prerequisites constraints and affiliated faculty

32.2. Flow of Events

32.2.1. Main Flow

1. The Administrator selects the “Manage Courses” option and chooses the “View Courses” function
2. The system retrieves the list of courses from the Microsoft SQL Server database
3. The system displays the course list with the key information of each course
4. The Administrator selects a course to view its details
5. The system retrieves the detailed information of the selected course from the Microsoft SQL Server database and displays it on the interface

32.2.2. Alternative Flow

A1: At Step 2, if the course list cannot be retrieved, the system displays an error message

A2: At Step 3, if no course exists in the system, the system displays a message stating that no courses were found

A3: At Step 5, if the detailed information of the selected course cannot be retrieved, the system displays an error message and returns to the course list

32.3. Special Requirements

The system shall respond to the view request within 15 seconds under normal operating conditions and restrict this function to the Administrator only

32.4. Preconditions

The user is authenticated as an Administrator and has an active session in the system

32.5. Postconditions

Success: The course list and the details of the selected course are displayed

Failure: An error message is displayed

32.6. Extension Points

None

33. Use-Case Update Courses

33.1. Summary

Allows the Administrator to update the information of an existing course, such as the course name, credit value, prerequisites constraints and affiliated faculty, while preventing changes to the identification data (course code)

33.2. Flow of Events

33.2.1. Main Flow

1. The Administrator selects the “Manage Courses” option and chooses the “Update Course” function on a selected course
2. The system retrieves the current information of the course from the Microsoft SQL Server database and displays it on the interface
3. The Administrator edits the course information and submits the request
4. The system validates the submitted information, including required fields, valid formats, that the credit value is a positive number, and that the selected prerequisites and faculty exist
5. The system updates the course record in the Microsoft SQL Server database

6. The system displays a success message and shows the updated course information

33.2.2. Alternative Flow

A1: At Step 2, if the course information cannot be retrieved, the system displays an error message and returns to the course list

A2: At Step 4, if the submitted information is invalid, incomplete, or the Administrator attempts to modify identification data, the system rejects the request, does not update the record and displays an error message identifying the invalid field

A3: At Step 4, if the selected prerequisites create a circular dependency (a course being its own prerequisite, directly or indirectly), the system rejects the request and displays an error message describing the conflict

A4: At Step 5, if the update cannot be committed to the database, the system keeps the course unchanged and displays an error message

33.3. Special Requirements

The system shall respond to the update request within 15 seconds under normal operating conditions, restrict this function to the Administrator only, not allow modification of the course code, and confirm completion only after the update is committed successfully to the database

33.4. Preconditions

The user is authenticated as an Administrator and has an active session in the system, and the selected course exists in the system

33.5. Postconditions

Success: The course information is updated in the system.

Failure: The course information remains unchanged and an error message is displayed

33.6. Extension Points

None

34. Use-Case Delete Courses

34.1. Summary

Allows the Administrator to delete a course, provided the course is not linked to ongoing class sections or historical academic records

34.2. Flow of Events

34.2.1. Main Flow

1. The Administrator selects the “Manage Courses” option and chooses the “Delete Course” function on a selected course
2. The system displays the details of the selected course
3. The system checks in the Microsoft SQL Server database whether the course is linked to ongoing class sections or historical academic records
4. If deletion is allowed, the system asks the Administrator for confirmation
5. The Administrator confirms the deletion
6. The system deletes the course record from the Microsoft SQL Server database
7. Upon successful database commitment, the system displays a success message and refreshes the course list

34.2.2. Alternative Flow

A1: At Step 3, if the course is linked to ongoing class sections or historical academic records, the system rejects the request, does not delete the course and displays the reason

A2: At Step 4 or 5, if the Administrator cancels the confirmation, the system does not delete the course and returns to the course list

A3: At Step 6, if the deletion cannot be committed to the database, the system rolls back all related changes, keeps the course unchanged and displays an error message

34.3. Special Requirements

The system shall respond to the delete request within 15 seconds under normal operating conditions, restrict this function to the Administrator only, require explicit

confirmation before deleting, and confirm completion only after the deletion is committed successfully to the database

34.4. Preconditions

The user is authenticated as an Administrator and has an active session in the system, and the selected course exists in the system

34.5. Postconditions

Success: The course is removed from the system and no longer appears in the course list

Failure: The course remains unchanged and an error message is displayed

34.6. Extension Points

None

35. Use-Case Manage Class Sections

35.1. Summary

Allows the Administrator to manage the class sections of a course within a specific semester. The assigned lecturer is set through Assign Lecturer to Class Sections

35.2. Flow of Events

35.2.1. Main Flow

1. The use case starts when the Administrator selects “Class section management”.
2. The Administrator selects a semester; the system displays the section list.
3. The Administrator chooses to create, view, update, or delete.
4. For View: The system displays the details of the selected class section: Class Code, Course Code, Semester, Assigned Lecturer, Maximum Capacity, Current Enrollment, Room and Class Schedule
5. For Create/Update: The Administrator enters Class Code, Course Code, Semester, Assigned Lecturer, Maximum Capacity, Room, and Class Schedule.

The system validates mandatory fields, the unique Class Code, the existing course and semester, and room and schedule conflicts, then saves to the database. Current Enrollment is maintained by the system and starts at 0.

6. For Delete: The system checks whether the class section can be deleted without violating data integrity constraints. If deletion is allowed, the system asks the Administrator for confirmation. After confirmation, the system deletes the class section and displays the result
7. Upon successful database commitment, the system displays a success message.

35.2.2. Alternative Flow

A1: At Step 2, if no class section exists or no class section matches the search criteria, the system displays a message stating that no class sections were found

A2: At Step 5, if a mandatory field is missing, the Class Code already exists, or the selected course or semester does not exist, the system rejects the request, performs rollback, and displays an error message identifying the invalid field

A3: At Step 5, if the room or class schedule conflicts with another class section, the system rejects the request, performs rollback, and displays an error message naming the conflicting room or schedule

A4: At Step 5, if the new Maximum Capacity is lower than the Current Enrollment, the system rejects the request and displays an error message identifying the invalid field

A5: At Step 6, if deletion violates data integrity constraints, the system rejects the request, keeps the class section unchanged, and displays the reason

A6: At Step 6, if the Administrator cancels the confirmation, the system cancels the operation and leaves database data unchanged

A7: At Step 5 or 6, if saving or deleting fails due to technical errors or system constraints, the system rolls back all related changes, keeps the class section unchanged, and displays an error message

35.3. Special Requirements

Only the Administrator is authorized to manage class sections. Each class section must belong to one course and one semester, and its Class Code must be unique. Current Enrollment must never exceed Maximum Capacity. Conflict errors must explicitly name the conflicting room or schedule

Response time for validating room/schedule conflicts and persisting section changes in the database must not exceed 15 seconds under standard test load conditions

35.4. Preconditions

The user is authenticated as an Administrator and has an active session in the system; for create/update, the semester and course exist; for view/update/delete, the class section exists

35.5. Postconditions

Success: Class section data is created, updated, or deleted in the database, or the class section details are displayed

Failure: Class section database records remain unchanged; error message is displayed.

35.6. Extension Points

Create Class Sections

View Class Sections

Update Class Sections

Delete Class Sections

Assign Lecturer to Class Sections

36. Use-Case Create Class Sections

36.1. Summary

Allows the Administrator to create a new class section for a course in a semester by entering its information, such as Class Code, Course Code, Semester, Maximum Capacity, Room and Class Schedule

36.2. Flow of Events

36.2.1. Main Flow

1. The Administrator selects the “Manage Class Sections” option and chooses the “Create Class Section” function
2. The system displays the class section creation form
3. The Administrator enters the class section information (Class Code, Course Code, Semester, Maximum Capacity, Room, Class Schedule) and submits the request
4. The system validates the submitted information, including required fields, valid formats, that the Class Code is unique, that the selected course and semester exist, and that the room and class schedule do not conflict with another class section
5. The system creates the new class section record in the Microsoft SQL Server database, with Current Enrollment set to 0 and no assigned lecturer
6. The system displays a success message and shows the newly created class section in the class section list

36.2.2. Alternative Flow

A1: At Step 4, if the submitted information is invalid or incomplete, the system rejects the request, does not create the class section and displays an error message identifying the invalid field

A2: At Step 4, if the Class Code already exists, the system rejects the request and displays an error message stating that the class section already exists

A3: At Step 4, if the room or class schedule conflicts with another class section, the system rejects the request, does not create the class section and displays an error message naming the conflicting room or schedule

A4: At Step 5, if the new record cannot be committed to the database, the system rolls back all related changes, creates no class section and displays an error message

36.3. Special Requirements

The system shall respond to the create request within 15 seconds under normal operating conditions, restrict this function to the Administrator only, display clear error messages for invalid inputs and schedule conflicts, and confirm completion only after the new class section is committed successfully to the database

36.4. Preconditions

The user is authenticated as an Administrator and has an active session in the system, and the course and semester for the class section already exist in the system

36.5. Postconditions

Success: The new class section is created and appears in the class section list

Failure: No class section is created and an error message is displayed

36.6. Extension Points

None

37. Use-Case View Class Sections

37.1. Summary

Allows the Administrator to view the list of class sections and the detailed information of a selected class section, including course, semester, lecturer, schedule, room, capacity and the number of enrolled students

37.2. Flow of Events

37.2.1. Main Flow

1. The Administrator selects the “Manage Class Sections” option and chooses the “View Class Sections” function
2. The system retrieves the list of class sections from the Microsoft SQL Server database
3. The system displays the class section list with the key information of each class section

4. The Administrator selects a class section to view its details
5. The system retrieves the detailed information of the selected class section from the Microsoft SQL Server database and displays it on the interface

37.2.2. Alternative Flow

A1: At Step 2, if the class section list cannot be retrieved, the system displays an error message

A2: At Step 3, if no class section exists in the system, the system displays a message stating that no class sections were found

A3: At Step 5, if the detailed information of the selected class section cannot be retrieved, the system displays an error message and returns to the class section list

37.3. Special Requirements

The system shall respond to the view request within 15 seconds under normal operating conditions and restrict this function to the Administrator only

37.4. Preconditions

The user is authenticated as an Administrator and has an active session in the system

37.5. Postconditions

Success: The class section list and the details of the selected class section are displayed

Failure: An error message is displayed

37.6. Extension Points

None

38. Use-Case Update Class Sections

38.1. Summary

Allows the Administrator to update the information of an existing class section, such as the schedule, room and maximum capacity, while preventing changes to the

identification data (class section code, course and semester). The assigned lecturer is changed through Assign Lecturer to Class Sections

38.2. Flow of Events

38.2.1. Main Flow

1. The Administrator selects the “Manage Class Sections” option and chooses the “Update Class Section” function on a selected class section
2. The system retrieves the current information of the class section from the Microsoft SQL Server database and displays it on the interface
3. The Administrator edits the class section information and submits the request
4. The system validates the submitted information, including required fields, valid formats, and that the new capacity is not lower than the current number of enrolled students
5. The system updates the class section record in the Microsoft SQL Server database
6. The system displays a success message and shows the updated class section information

38.2.2. Alternative Flow

A1: At Step 2, if the class section information cannot be retrieved, the system displays an error message and returns to the class section list

A2: At Step 4, if the submitted information is invalid, incomplete, or the Administrator attempts to modify identification data, the system rejects the request, does not update the record and displays an error message identifying the invalid field

A3: At Step 4, if the new schedule or room conflicts with an existing class section, the system rejects the request and displays an error message describing the conflict

A4: At Step 5, if the update cannot be committed to the database, the system keeps the class section unchanged and displays an error message

38.3. Special Requirements

The system shall respond to the update request within 15 seconds under normal operating conditions, restrict this function to the Administrator only, not allow modification of the class section code, course or semester, and confirm completion only after the update is committed successfully to the database

38.4. Preconditions

The user is authenticated as an Administrator and has an active session in the system, and the selected class section exists in the system

38.5. Postconditions

Success: The class section information is updated in the system

Failure: The class section information remains unchanged and an error message is displayed

38.6. Extension Points

None

39. Use-Case Delete Class Sections

39.1. Summary

Allows the Administrator to delete a class section, provided the deletion does not violate data integrity constraints

39.2. Flow of Events

39.2.1. Main Flow

1. The Administrator selects the “Manage Class Sections” option and chooses the “Delete Class Section” function on a selected class section
2. The system displays the details of the selected class section
3. The system checks in the Microsoft SQL Server database whether the class section can be deleted without violating data integrity constraints
4. If deletion is allowed, the system asks the Administrator for confirmation

5. The Administrator confirms the deletion
6. The system deletes the class section record from the Microsoft SQL Server database
7. Upon successful database commitment, the system displays a success message and refreshes the class section list

39.2.2. Alternative Flow

A1: At Step 3, if deletion violates data integrity constraints, the system rejects the request, does not delete the class section and displays the reason

A2: At Step 4 or 5, if the Administrator cancels the confirmation, the system does not delete the class section and returns to the class section list

A3: At Step 6, if the deletion cannot be committed to the database, the system rolls back all related changes, keeps the class section unchanged and displays an error message

39.3. Special Requirements

The system shall respond to the delete request within 15 seconds under normal operating conditions, restrict this function to the Administrator only, require explicit confirmation before deleting, and confirm completion only after the deletion is committed successfully to the database

39.4. Preconditions

The user is authenticated as an Administrator and has an active session in the system, and the selected class section exists in the system

39.5. Postconditions

Success: The class section is removed from the system and no longer appears in the class section list

Failure: The class section remains unchanged and an error message is displayed

39.6. Extension Points

None

40. Use-Case Assign Lecturer to Class Sections

40.1. Summary

Allows the Administrator to assign a lecturer to a specific class section in a semester.

40.2. Flow of Events

40.2.1. Main Flow

1. The use case starts when the Administrator selects “Assign lecturer”.
2. The Administrator selects a semester and class section.
3. The system verifies section availability and displays the lecturer list.
4. The Administrator selects a lecturer and clicks Confirm.
5. The system checks the database for teaching schedule conflicts for the selected lecturer.
6. The system saves the assignment to the database.
7. Upon successful database commitment, the system displays a success message.

40.2.2. Alternative Flow

A1: At Step 5, if the lecturer has a schedule conflict with another class section in the semester, the system rejects the assignment and displays an error message naming the conflicting class section and schedule

A2: At Step 6, if saving fails, the system rolls back the transaction, keeps the assignment unchanged and displays an error message

40.3. Special Requirements

Only the Administrator is authorized to assign lecturers

Response time for checking schedule conflicts and committing assignments to the database must not exceed 15 seconds under standard test load conditions

40.4. Preconditions

The user is authenticated as an Administrator and has an active session in the system; the class section and the lecturer exist

40.5. Postconditions

Success: The assignment is saved in the database; the lecturer gains authority to manage grades for the section.

Failure: Assignment status remains unchanged in the database; error message is displayed.

40.6. Extension Points

None

CHAPTER 5: CLASS DIAGRAM

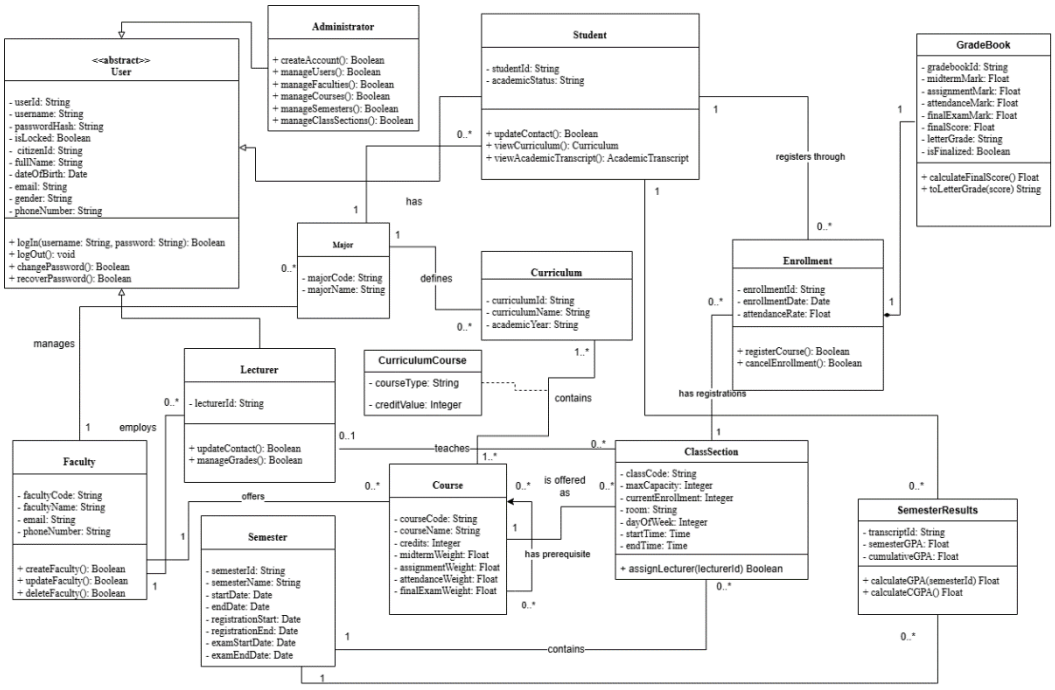


Image 4: Class Diagram

CHAPTER 6: DATA MODEL

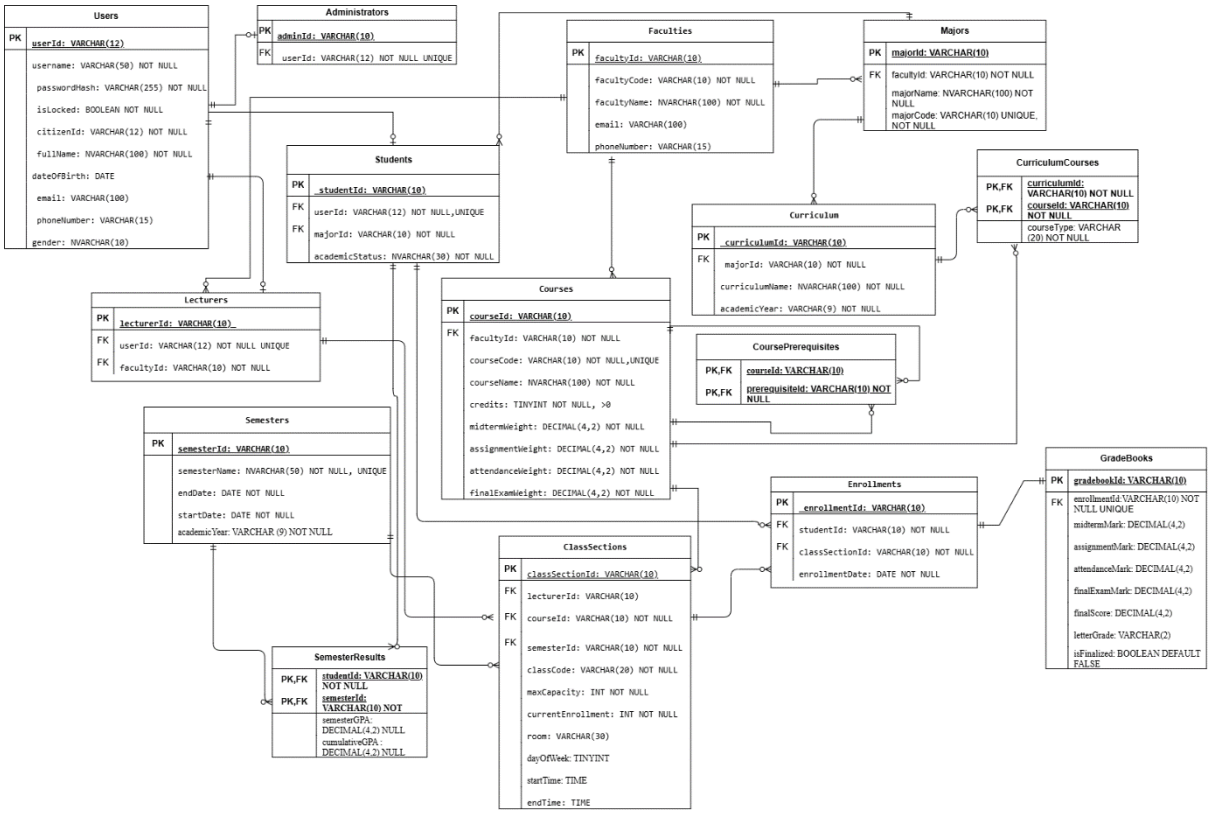


Image 5: Data Model

CHAPTER 7: INTERFACE DESIGN DESCRIPTION

1. Overall Interface Overview

The Student Management System (SMS) is a web-based application with a responsive graphical user interface (GUI). Figma is used to design the interfaces. In the next phase, the interface will be built with HTML, CSS, and JavaScript; Python will handle the main functions, and Microsoft SQL Server will store the data. The system will be deployed on a Windows 11 server

Users access the system through a web browser and work with visual forms, tables, dashboards, and navigation menus instead of commands. Based on user roles (admin, student, lecturer), the system follows role-based access control (RBAC) to provide separate menus with appropriate functions

2. Authentication Design

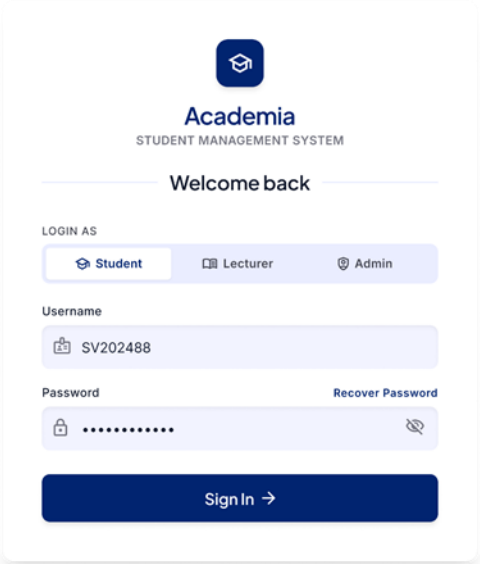


Image 6: Login Interface

The login page is the entry point for all users. It provides the following options:

- Select a role (student, lecturer, or admin), then log in using a username and password. The password is masked and can be shown or hidden. The selected role is only used to check the account. After a successful login, the system opens the interface of the role saved in the account.
- If the username or password is incorrect, the selected role does not match the account, or the account is locked, the system displays an appropriate error message without revealing sensitive account information. A role mismatch shows the same message as incorrect login information.
- Recover password through the registered email or phone number. After verifying the account, the user sets a new password directly on the screen.
- After 15 minutes of inactivity, the session ends and the user returns to the login page with the message “Session expired due to inactivity.”
- After logging in, users can log out from any screen.

3. Role-Based Menu Design

Each role is provided with a distinct set of functions

3.1. Admin Interface

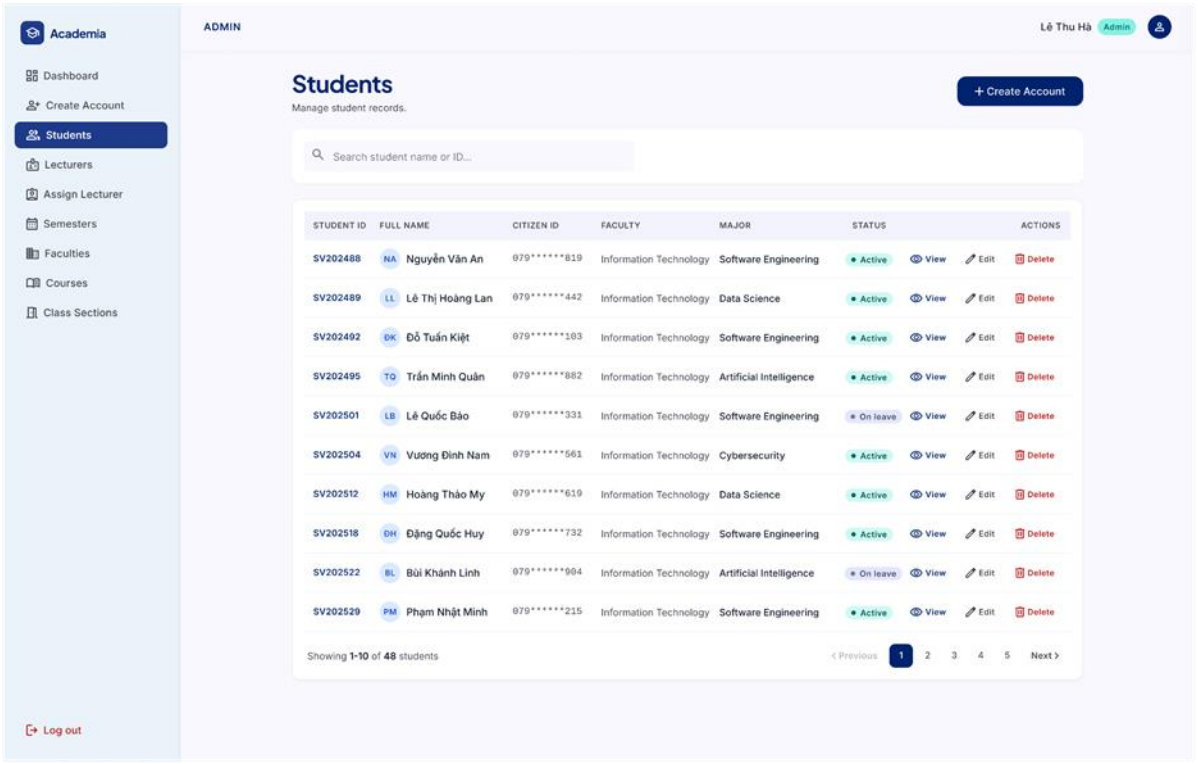


Image 7: Administrator Interface

The admin interface allows admins to manage user accounts and academic data. Its sidebar includes Dashboard, Create Account, Students, Lecturers, Assign Lecturer, Semesters, Faculties, Courses, Class Sections, and Log out. Main functions include:

- Creating student and lecturer accounts through Create Account. IDs are generated automatically. The “Create Account” button on the Students page opens the same form for the Student account type.
- Managing student records and lecturer profiles (view, update, delete) and searching by name or ID. Deleting a record requires a data check and confirmation.
- Assigning lecturers to class sections. The system checks for schedule conflicts.
- Managing Faculties, Courses, and Class Sections (view, create, update, delete).

- Managing Semesters (create, view, configure, delete): dates, registration period, and exam schedule.
- Logging out of the system. Admins have no profile or change-password function.

3.2. Student Interface

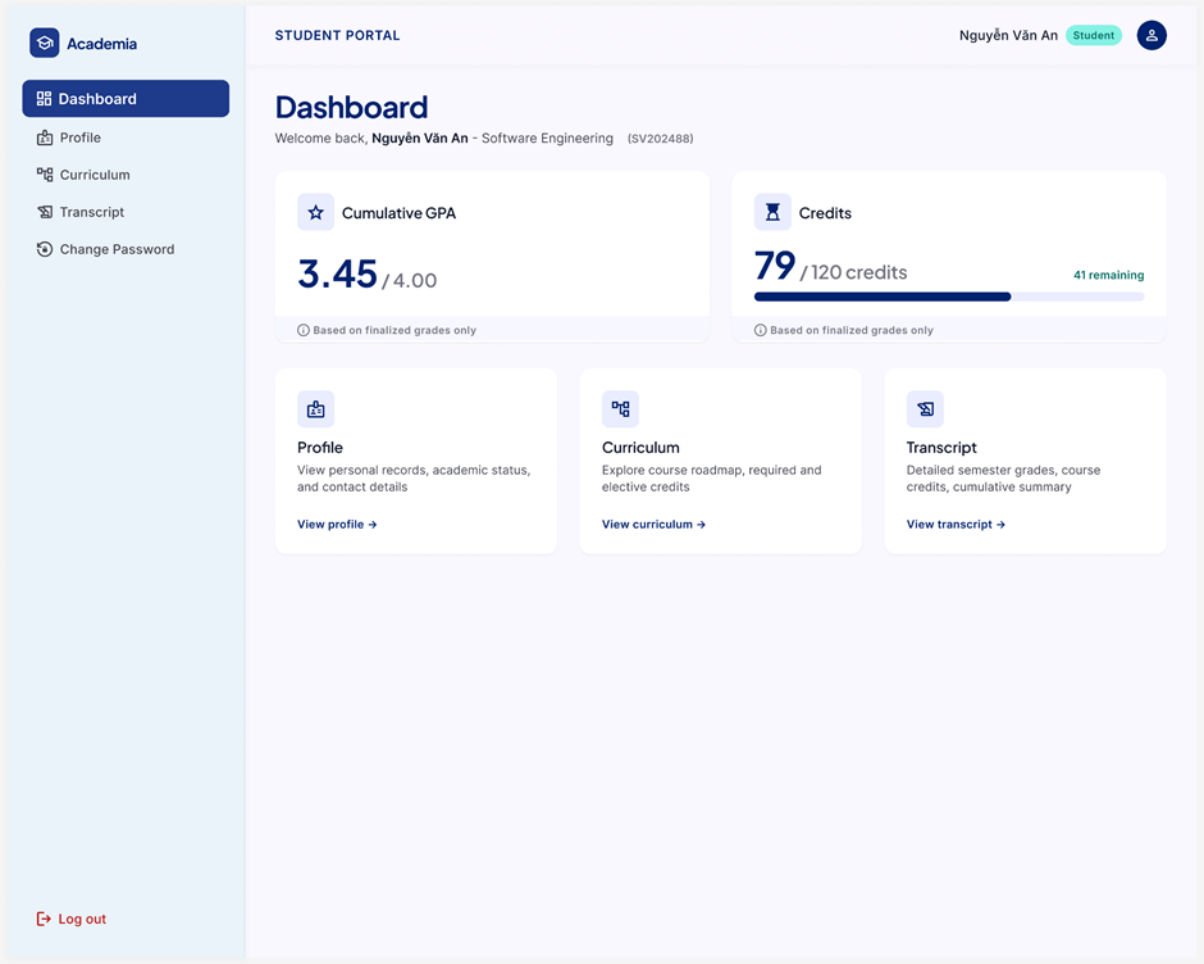


Image 8: Student Interface

The student interface supports academic tracking. Its sidebar includes Dashboard, Profile, Curriculum, Transcript, Change Password, and Log out. Students can perform the following actions:

- Viewing the dashboard, which displays the Cumulative GPA and the number of completed credits based on finalized grades only.
- Viewing their profile and updating contact information (email, phone number).
- Viewing the curriculum of their major, with each course labeled “Completed” or “Not Completed."
- Viewing the transcript, including finalized marks, letter grades, semester GPA, and cumulative GPA.
- Changing their password and logging out of the system

3.3. Lecturer Interface

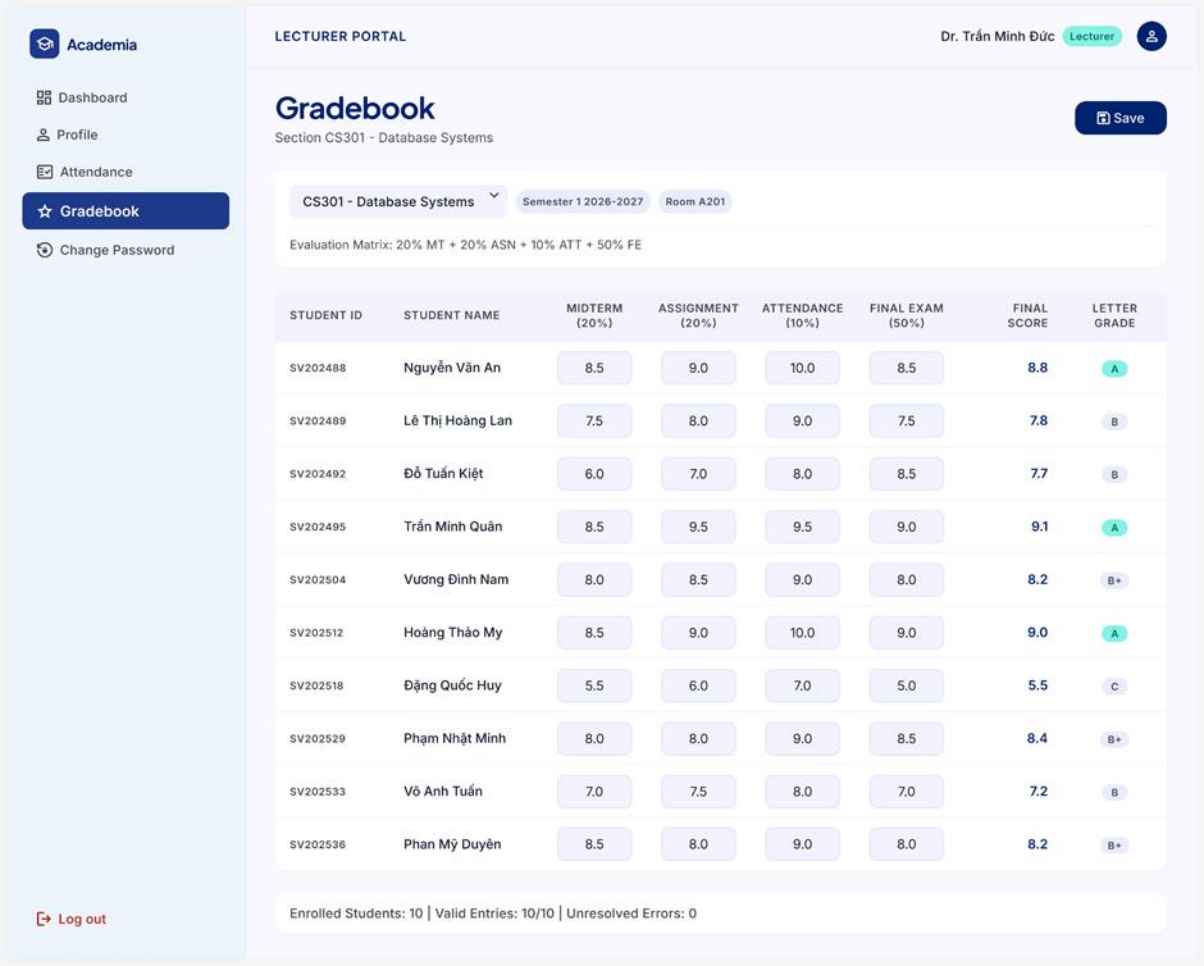


Image 9: Lecturer Interface

The lecturer interface provides teaching-related functions. Its sidebar includes Dashboard, Profile, Attendance, Gradebook, Change Password, and Log out. Lecturers can:

- View their profiles and update contact information, excluding their faculty.
- Enter component marks (0.0-10.0) for their assigned class sections. The final grade and letter grade are calculated automatically.
- Change their passwords and log out of the system.

4. User Interaction and System Feedback

The system provides immediate and clear feedback after each user action.

- If an operation is successful, the system displays a success message. Important actions, such as changing a password or deleting a record, require confirmation.
- If an operation fails, the system displays an error message. The system prevents incomplete changes from being saved, and the original data remains unchanged.
- Error messages are clear and highlight the field causing the error. Grades must be between 0.0 and 10.0."
- If a student or lecturer tries to change identification data or faculty, the system rejects the update. Only contact information can be changed.

5. Interface Design Rationale

A web-based GUI was selected for this system due to the following reasons:

- The non-functional requirements state that the interface must be easy to use, consistent, and responsive, with feedback after each action.
- Figma allows the group to design and prototype the interfaces before coding.
- Separate menus for each role reduce mistakes and support access control.

- A graphical interface can highlight the fields that cause errors.
- Tables and status labels display curriculum, transcripts, and gradebooks clearly.